

The conjectured sharp multiplier exponent for a flat zero patch

Candidate substantial partial-result packet for arXiv:1907.08812

11 August 2026

Status: candidate substantial partial result, likely valid, subject to human review.
For $1/2 < s < 1$, if the Sobolev trace of $w \in H^s(\mathbb{T}^d)$ vanishes on a nonempty open patch of a rational affine hyperplane, then $1/w \notin \mathcal{M}_2^q$ for every $2 \leq q \leq 1/(1-s)$. This is exactly the exponent conjectured by the source when the zero-set dimension is $\sigma = d-1$.

1 Source question and scope

Northington studies reciprocals $u = 1/w$ as $(2, q)$ Fourier multipliers and proves that if $w \in H^s(\mathbb{T}^d)$ has a zero set of positive σ -dimensional Hausdorff measure, then $u \notin \mathcal{M}_2^q$ in a stated range. The paper asks whether this range can be sharpened to

$$q \leq \frac{d - \sigma}{d - \sigma - s}. \quad (1)$$

See Section 7, Question 1, page 30 of the source [1]; the page is reproduced in Appendix A.

The theorem below establishes (1) for $\sigma = d-1$ under the stronger geometric assumption that the zero set contains an open flat patch in the Sobolev-trace sense. It does not treat arbitrary fractal zero sets of positive Hausdorff measure.

We use the source's equivalent formulation of the multiplier condition: if $u = 1/w \in \mathcal{M}_2^q$, then for every measurable g with $wg \in L^2$,

$$\|\widehat{g}\|_{\ell^q(\mathbb{Z}^d)} \leq C \|wg\|_{L^2(\mathbb{T}^d)}. \quad (2)$$

Idea of the proof

Near a flat zero patch, test (2) on a slab of normal thickness τ and fixed tangential footprint. Its Fourier transform factors, and its ℓ^q norm is bounded below by $\tau^{1-1/q}$. Since the normal one-dimensional sections of w vanish at the middle of the slab, a scaled fractional Poincaré inequality bounds the other side by $\tau^s E(\tau)$, where $E(\tau) \rightarrow 0$ is the Sobolev energy in the shrinking slab. At $q = 1/(1-s)$ the powers of τ exactly tie, and the vanishing energy produces the endpoint contradiction. A unimodular torus change of variables gives every rational affine hyperplane.

2 Two local estimates

Write points of $\mathbb{T}^d$ as $(t, y) \in \mathbb{T} \times \mathbb{T}^{d-1}$.

Lemma 1 (interval Fourier mass). *For $2 \leq q < \infty$ and $0 < \tau < 1/8$,*

$$\|\widehat{\mathbf{1}_{(-\tau, \tau)}}\|_{\ell^q(\mathbb{Z})} \geq c_q \tau^{1-1/q}.$$

Proof. For every integer k with $|k| \leq (8\tau)^{-1}$,

$$\left| \int_{-\tau}^{\tau} e^{-2\pi i k t} dt \right| = \left| \frac{\sin(2\pi k \tau)}{\pi k} \right| \geq c\tau,$$

with the value 2τ at $k = 0$. There are at least c/τ such integers, which gives the claim after taking the ℓ^q norm. $\square$

Lemma 2 (localized anchored fractional Poincaré). *Let $1/2 < s < 1$. If $f \in H^s(-2\tau, 2\tau)$ has its continuous representative satisfying $f(0) = 0$, then*

$$\int_{-\tau}^{\tau} |f(t)|^2 dt \leq C_s \tau^{2s} \int_{-2\tau}^{2\tau} \int_{-2\tau}^{2\tau} \frac{|f(t) - f(r)|^2}{|t - r|^{1+2s}} dr dt.$$

Proof. After scaling, this is the standard one-dimensional fractional Sobolev estimate

$$|F(x) - F(0)| \leq C_s |x|^{s-1/2} [F]_{H^s(-2,2)}, \quad |x| \leq 1,$$

integrated over $x \in (-1, 1)$. The point value is legitimate because $s > 1/2$. $\square$

3 Sharp exponent on a flat patch

Theorem 1. *Let $d \geq 2$, $1/2 < s < 1$, and $w \in H^s(\mathbb{T}^d)$. Suppose there is a nonempty open set $U \subset \mathbb{T}^{d-1}$ such that the Sobolev trace of w on $\{0\} \times \mathbb{T}^{d-1}$ vanishes almost everywhere on U . Then*

$$\frac{1}{w} \notin \mathcal{M}_2^q(\mathbb{T}^d) \quad \text{for every} \quad 2 \leq q \leq \frac{1}{1-s}.$$

Proof. Choose a nonzero $\phi \in C_c^\infty(U)$. The Fourier characterization of $H^s(\mathbb{T}^d)$ and Fubini give

$$\int_{\mathbb{T}^{d-1}} [w(\cdot, y)]_{H^s(\mathbb{T})}^2 dy \asymp \sum_{(k_1, k') \in \mathbb{Z}^d} |k_1|^{2s} |\widehat{w}(k_1, k')|^2 \leq \|w\|_{H^s(\mathbb{T}^d)}^2. \quad (3)$$

Thus almost every normal section belongs to $H^s(\mathbb{T})$, its value at zero equals the Sobolev trace, and Lemma 2 applies on the support of ϕ .

For small $\tau > 0$, put

$$g_\tau(t, y) = \mathbf{1}_{(-\tau, \tau)}(t) \phi(y).$$

The Fourier transform factors. Lemma 1 therefore gives

$$\|\widehat{g}_\tau\|_{\ell^q(\mathbb{Z}^d)} = \|\widehat{\mathbf{1}_{(-\tau, \tau)}}\|_{\ell^q(\mathbb{Z})} \|\widehat{\phi}\|_{\ell^q(\mathbb{Z}^{d-1})} \geq c_{q, \phi} \tau^{1-1/q}. \quad (4)$$

Define the local sectional energy

$$E(\tau)^2 = \int_{\text{supp } \phi} \int_{-2\tau}^{2\tau} \int_{-2\tau}^{2\tau} \frac{|w(t, y) - w(r, y)|^2}{|t - r|^{1+2s}} dr dt dy.$$

By (3), the full integrand is integrable. Absolute continuity of the integral gives $E(\tau) \rightarrow 0$ as $\tau \downarrow 0$. Integrating Lemma 2 in y yields

$$\|wg_\tau\|_2 \leq C_{s, \phi} \tau^s E(\tau). \quad (5)$$

Assume, for contradiction, that $1/w \in \mathcal{M}_2^q$; in particular w is nonzero almost everywhere, so the source's testing identity applies without ambiguity. Apply (2) to g_τ and combine (4)–(5):

$$\tau^{1-1/q-s} \leq CE(\tau).$$

If $q < 1/(1-s)$, the exponent on the left is negative, so the left side diverges. At the critical value $q = 1/(1-s)$, the left side is one while $E(\tau) \rightarrow 0$. Both cases are impossible. $\square$

Corollary 1 (rational affine hyperplanes). *Let $a \in \mathbb{Z}^d$ be primitive and let $H_{a,c} = \{x \in \mathbb{T}^d : a \cdot x = c \pmod{1}\}$. If the Sobolev trace of $w \in H^s(\mathbb{T}^d)$ vanishes on a nonempty relatively open patch of $H_{a,c}$, then the conclusion of Theorem 1 holds.*

Proof. Extend a to a unimodular integer matrix. The resulting torus automorphism, followed by a translation, sends $H_{a,c}$ to a coordinate hyperplane. Composition preserves Lebesgue norms, permutes Fourier coefficients by a lattice bijection, preserves the $(2, q)$ multiplier property, and acts boundedly and invertibly on H^s . Apply Theorem 1. $\square$

4 Verification, limitations, and novelty

The proof was audited at the two potentially sharp points. First, the slab Fourier lower bound counts $\asymp \tau^{-1}$ coefficients of size $\asymp \tau$. Second, the endpoint does not follow from a global Poincaré constant: it specifically uses absolute continuity of the *localized* sectional energy, which tends to zero. The assumption $s > 1/2$ is exactly what makes the normal trace and anchored one-dimensional inequality available.

This packet proves the conjectured exponent only for a flat trace-zero patch. It does not settle arbitrary sets with $\mathcal{H}^{d-1}(\Sigma(w)) > 0$, lower-dimensional or fractal zero sets, the $s = 1$ boundary, or the other open problems in Section 7.

The bounded novelty check on 11 August 2026 searched the run registry and all four cheap indexes for the arXiv id, title, sharp exponent, hyperplane, and zero-set phrases. Exact-title and close-phrase primary-source web searches found the source and unrelated multiplier papers, but no later resolution or this flat-patch theorem. This is not an exhaustive priority claim.

Human-review recommendation. Check the localized anchored fractional Poincaré estimate and the identification of the sectional value at $t = 0$ with the Sobolev trace. These are standard for $s > 1/2$ and are the only functional-analytic inputs beyond the source’s multiplier testing identity.

A Source-page evidence

References

- [1] M. Northington, *Uncertainty Principles for Fourier Multipliers*, arXiv:1907.08812 [math.FA], 2019.

that is, suppose that $\hat{F} \in [W^{s,2}(\mathbb{R}^d)]^K$ for $s = \min(d/q' - d/2 + 1/2, 1)$. By Proposition 6.4 it follows that the eigenvalues ζ_i of $P := P(\hat{F})$ satisfy $\sqrt{\zeta_i} \in W^{s,2}(\mathbb{T}^d)$ for all $i \in \{1, \dots, K\}$. On the other hand, Proposition 6.7 implies that $\mathcal{H}^{d-1}(\Sigma(\det(P^{1/2}))) > 0$. Finally, since $\mathcal{T}(F)$ is a minimal (C_q) -system, Proposition 6.3 implies that $P^{-1/2} \in \mathfrak{M}_2^q(K)$. So, by corollary 1.10 we must have $s > \min(d/q' - d/2 + 1/2, 1)$, which is a contradiction.

We now turn to prove the sharpness of Theorem 1.13. For $2 < q \leq \infty$, let $\beta < 1 - \frac{2}{q}$, and let f_β be the function defined in Proposition 2.13. Then, in particular, f_β satisfies (45) for all $t < 1 + \beta$. Note that taking β arbitrarily close to $1 - \frac{2}{q}$, allows $1 + \beta$ to become arbitrarily close to the critical exponent $2/q'$. Consider the family F containing the single function $F = \{f_\beta\}$ and note that $P(\hat{f}_\beta)$ is a scalar valued function which is equal to $(\hat{f}_\beta)^2$ on $[-\frac{1}{2}, \frac{1}{2}]$. By applying the Fourier transform it is readily checked that $V(f_\beta)$ is translation invariant, that is, that it is invariant under any real shift.

We claim that $P(\hat{f}_\beta)^{-1/2} \in \mathcal{M}_2^q$, so that $\mathcal{T}(f_\beta)$ is an exact (C_q) -system for $V(f_\beta)$. Indeed, by the discussion following Theorem 1.5 it is enough to check that $P(\hat{f}_\beta)^{-1/2} \in L^{\frac{2q}{q-2}}(\mathbb{T})$. We have

$$\int_{\mathbb{T}} P(\hat{f}_\beta)(\xi)^{-\frac{q}{q-2}} dx = 2 \int_0^{1/2} \left(\frac{1}{2} - \xi\right)^{-\frac{q}{q-2}} dx,$$

and this integral is finite since $\beta < 1 - \frac{2}{q}$.

Finally, we use the construction above to demonstrate the sharpness of the case $q = \infty$ in Theorem 1.14. For any $d \in \mathbb{N}$ and $\beta < 1$, let $F_\beta(x) = f_\beta(x_1)f_\beta(x_2) \cdots f_\beta(x_d)$. Then, by Proposition 2.13, F_β satisfies (45) for all $t < 2$. Consider the family F containing the single function $F = \{F_\beta\}$ and note that as above $P(\hat{F}_\beta)$ is a scalar valued function which is equal to $(\hat{F}_\beta)^2$ on $[-\frac{1}{2}, \frac{1}{2}]^d$. As above, $V(F_\beta)$ is translation invariant, that is, that it is invariant under any real shift. Since $P(\hat{F}_\beta)^{-1/2} \in L^2(\mathbb{T}^d) = \mathcal{M}_2^\infty$, it follows that $\mathcal{T}(F_\beta)$ is an exact (C_∞) -system for $V(F_\beta)$. The proof is complete. $\square$

7. FURTHER DIRECTIONS OF STUDY AND OPEN PROBLEMS

In this section we outline some questions related to this work.

1. As mentioned in the introduction, it is likely that the results in theorems 1.5 and 1.14 are not sharp (see also Theorem 3.3 and its corollaries). Following the approach developed above, any refinement of Theorem 1.5 will imply in turn an improvement of the result in Theorem 1.14. In particular, as discussed in the introduction, it is natural to ask whether the upper bound in part i of Theorem 1.5 can be replaced by the bound $q \leq (d - \sigma)/(d - \sigma - s)$. If such a result holds then Theorem 1.14 will remain true with the bound $t \geq 2/q'$. With a similar argument to that of the sharpness of Theorem 1.13 one can show that no bigger bound is possible in Theorem 1.14.
2. The parameters of the Sobolev space $W^{s,r}(\mathbb{T}^d)$ considered in this paper are restricted to satisfy $\alpha = s - d/r < 1$, which implies that this space is embedded into the space of α -Hölder continuous functions. To obtain sharp bounds in the case of $\alpha = 1$, some refinement of the results is expected. In particular, such a refinement in Theorem 2.11 will allow to extend Theorem 5.3 to the full boundary of the region described in [36].

Figure 1: Page 30 of arXiv:1907.08812. Question 1 asks whether the range can be sharpened to $q \leq (d - \sigma)/(d - \sigma - s)$.